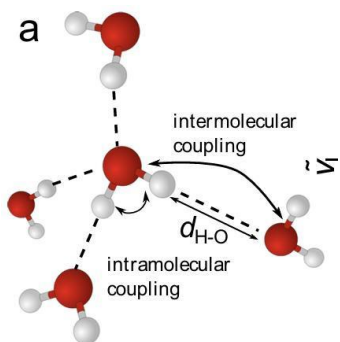


Background & Goals

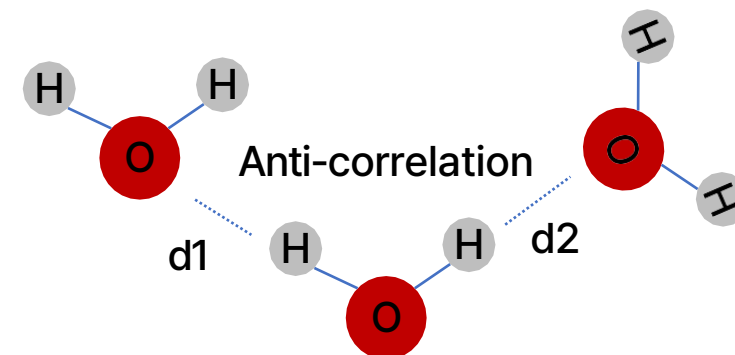
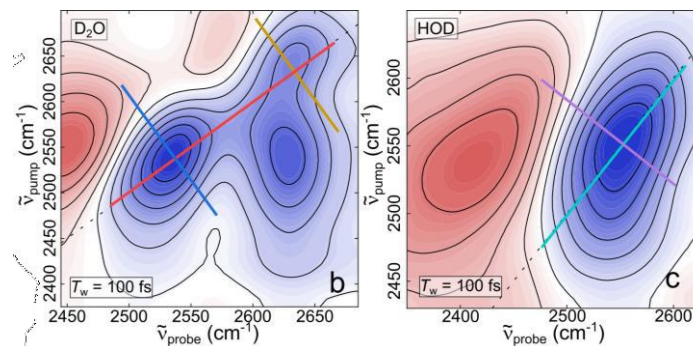
Hydrogen Bond Network



- Water has anomalous physical properties attributed to its hydrogen-bond network.
- However, the conventional **tetrahedral picture** is insufficient to describe its **local structural heterogeneity**.
- Two O-H bonds in a single molecule can form **unequal hydrogen bonds**, leading to **structural asymmetry**.

Anti-correlation of Liquid Water

Our group observed **anti-correlation between two O-H stretch modes** using 2D-IR, suggesting intramolecular asymmetry



Yet, their origin and structural distribution remain unclear.

Research Questions

Q1. What is the origin of hydrogen-bond asymmetry or anti-correlation in water? (Thermal Factor?)

Q2. Can we prove anti-correlation in simulation?

Specifically:

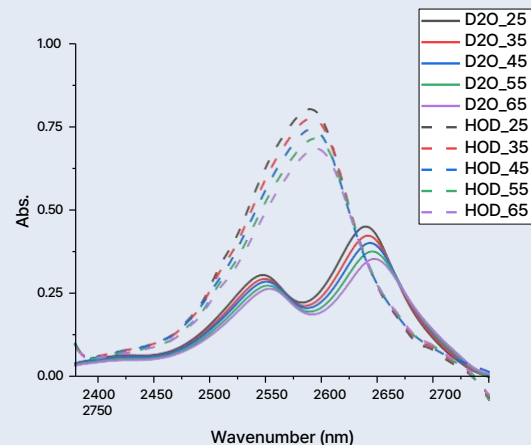
- How does temperature affect hydrogen–bond heterogeneity?
- Can anti–correlation be reproduced and quantified in simulations?
- What is the relationship between asymmetry and anti–correlation?

Approach

Combined temperature-dependent FTIR spectroscopy with molecular simulations to probe hydrogen-bond network structures and vibrational behavior.

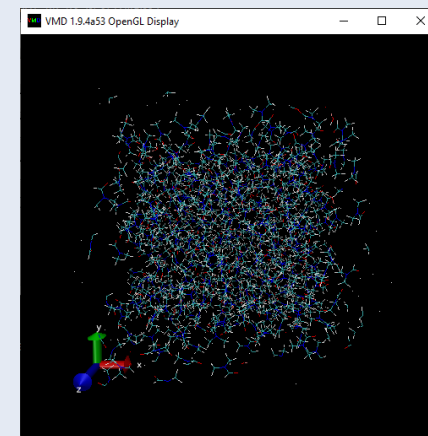
Experiment – FTIR

- Temperature-dependent FTIR measurements (25–65 °C)
- HOD and D₂O diluted in DMF were used to:
 - suppress intermolecular coupling
 - isolate intramolecular vibrational behavior
- The O–D stretching band ($\sim 2500 \text{ cm}^{-1}$) was analyzed.



Simulation – MD simulation

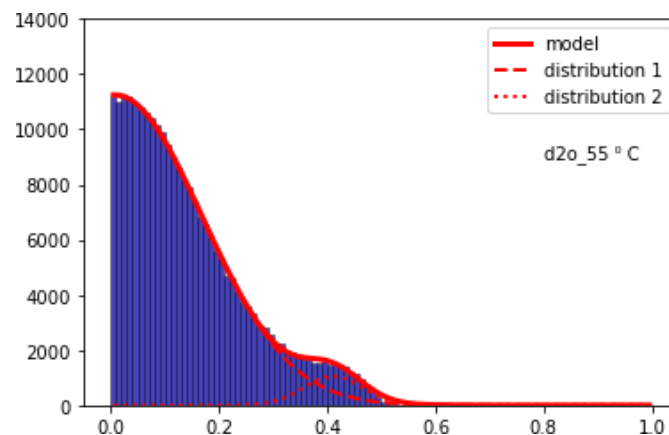
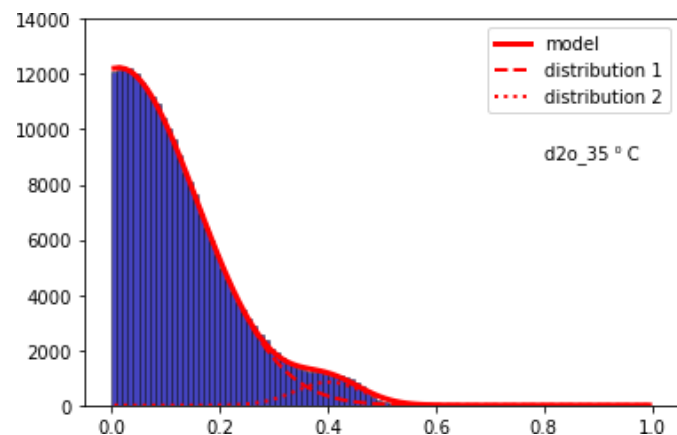
- Classical molecular dynamics simulations to model water structure.
- Two water models were used:
 - SPC/E (non-polarizable)
 - SWM4-NDP (polarizable)
- Analysis:
 - # of HB
 - HB distances (d_{HB})
 - Radial distribution function
 - Time correlation function and VDOS



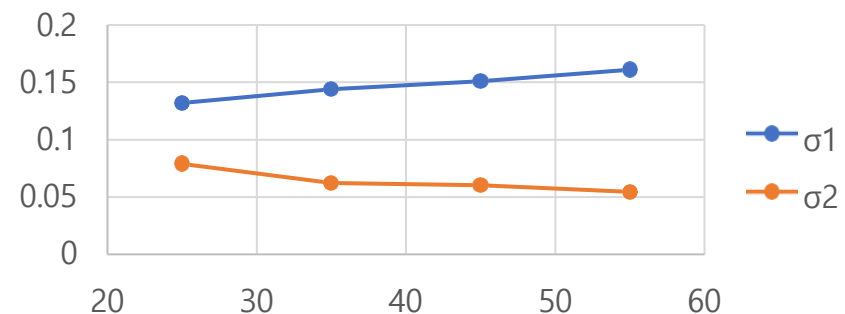
Results

Asymmetry Distribution

Define Asymmetry Parameter $\gamma = 1 - \frac{\min(d1,d2)}{\max(d1,d2)}$



Gaussian Fit with $f = A \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

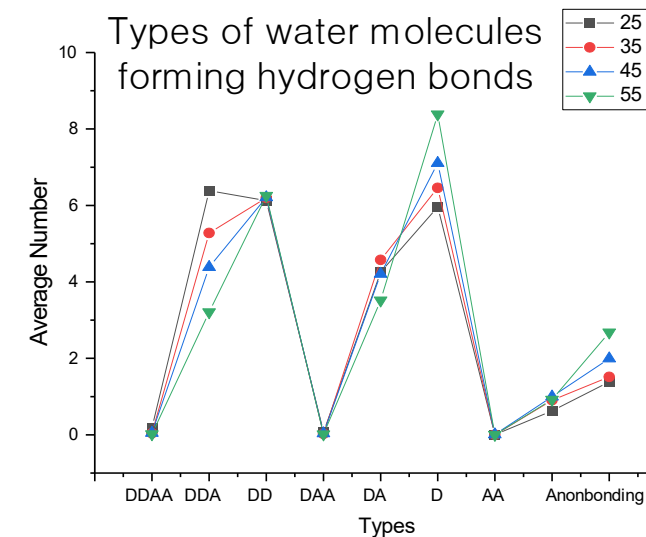
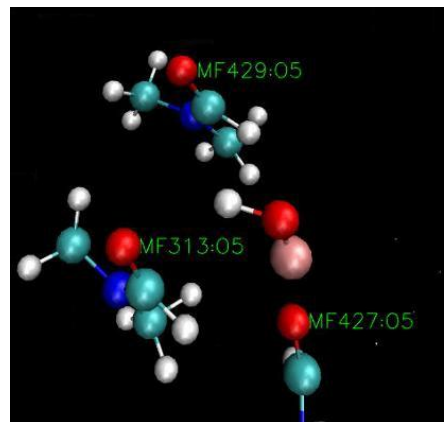
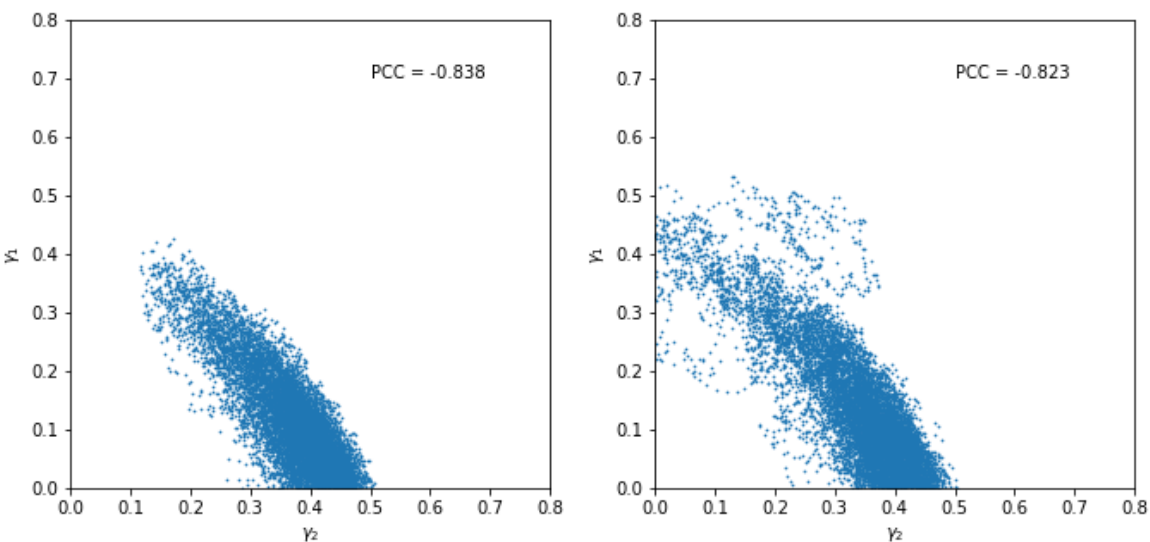


- Additional **distinct high-asymmetry regime** emerges.
- Bimodal Gaussian fit show that the main distribution broadens with temperature, while the high-asymmetry component becomes narrower but increases in contribution.
- ➔ Increasing **thermal energy** does not only induce random fluctuations, but also enhances **the population of a structurally distinct asymmetric hydrogen-bonding state**.
- Based on this observation, let's further investigated its possible connection to the anti-correlation behavior.

Results

Non-bound Water molecules

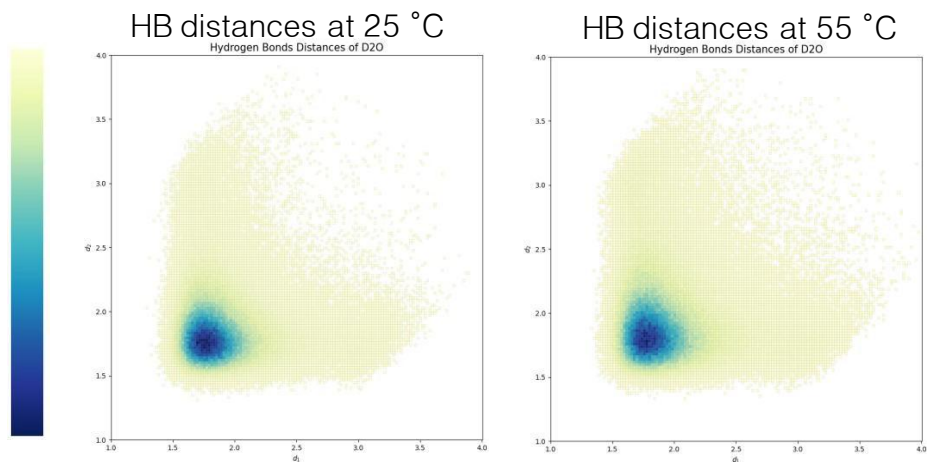
Define γ_2 : asymmetry parameter of nearest and 2nd nearest interaction distances of the less bound hydrogen



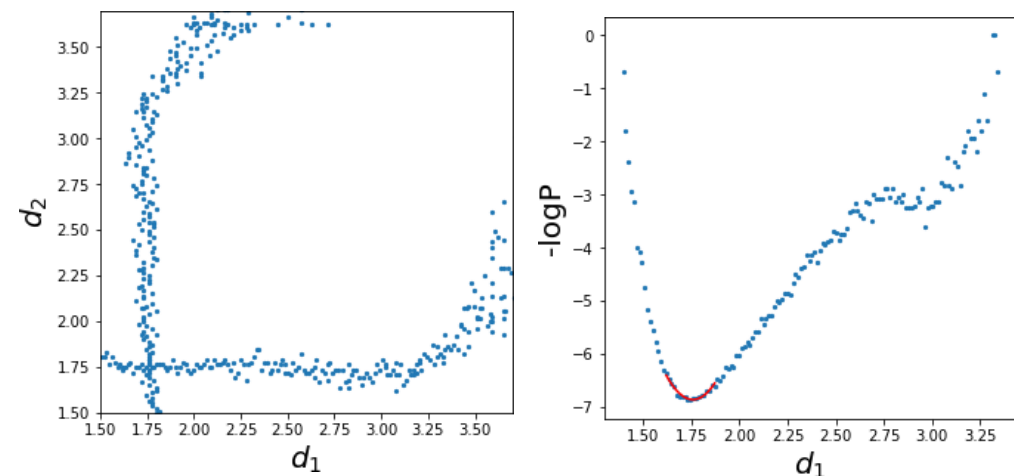
- Significant negative correlation between γ_1 and γ_2 .
- Appearance of distinctly asymmetric state attributed to [competing interaction](#), which makes “non-bound water” of “non-bonding OH group.”
- Temperature $\uparrow \rightarrow$ # of non-bound water molecules $\uparrow$.
- Two HB distances have positive correlation when they are highly asymmetric.

Results

Anti-correlation

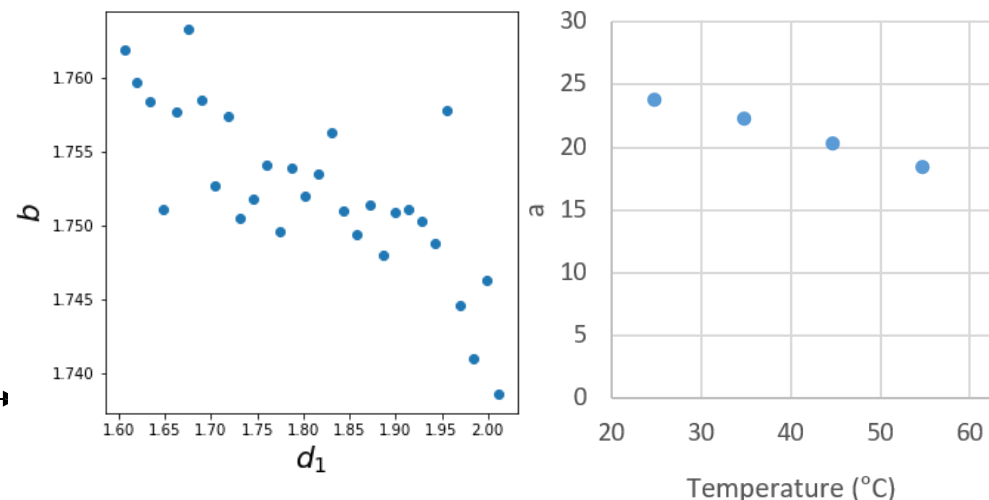


- Analyzed the conditional free energy landscape via $F(d_2 | d_1) = -\log P(d_2 | d_1)$ with the HB distribution as shown.
 - Anti-correlation between d_1 and d_2 is observed across all temperatures, while the strength of coupling decreases as temperature increases.
- ➔ Thermal fluctuations partially weaken the effective coupling, while the anti-correlation is an intrinsic structural feature of liquid water rather than a purely thermal effect.



Fit the potential curve around minimum with harmonic approximation

$$F(d_1) = a(x - b)^2 + c$$



Conclusion & Insights

- Temperature-dependent FTIR and MD simulations revealed progressive weakening and reorganization of the HB network.
- The anti-correlation behavior was reproduced in MD simulations.
- Anti-correlation is an **intrinsic dynamic phenomenon** rather than a simple statistical or thermal population effect in contrast to static structural asymmetry.
- Thermal fluctuation promoted the formation of **non-bound water populations (which led to the change in vibrational spectrum)** and weakened intermolecular vibrational coupling, leading to a more heterogeneous and weakly correlated vibrational ensemble.
- Vibrational responses in water are governed by collective intermolecular coupling dynamics as well as local HB strength.